

GOAD v3 — Simple Install— Ubuntu & Troubleshooting

 medium.com/@shanksf/goad-v3-simple-install-ubuntu-troubleshooting-7505de7e7322

Shanks

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GOAD v3

The new version of **GOAD** is out and it's better than v2, so I'll show you how to get the most out of it **first try**, without **no(t)** (much) troubleshooting.

This was fully tested only on a Windows 10 machine with **64GB of RAM**, using **VMware** and **Ubuntu**.

This may also work for GOAD-light, GOAD-Mini, SSCM, NHA and so on.

It's the easiest guide on the internet for using and pwning GOAD. If you wanna use the v2, here's the tutorial for [WINDOWS](#).

Leave a comment if works for you. :)

| Main project by Mayfly: <https://orange-cyberdefense.github.io/GOAD/>

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PREREQUISITES

- VMWARE Workstation (Also may work with Virtualbox, just adapt)
- Kali or Ubuntu (Choose any distro you want, I'll use Ubuntu just for demonstration)
- A lot of disk space (~120GB)
- A lot of ram (At least 32GB for the main GOAD)

STEPS

After installing and setting up Ubuntu just run the following:

```
# Install vbox
sudo apt install virtualbox
```

```
# Install vagrant
wget -O- https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/hashicorp-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | sudo tee
/etc/apt/sources.list.d/hashicorp.list
sudo apt update && sudo apt install vagrant
```

```
# Install Vagrant plugins
vagrant plugin install vagrant-reload vagrant-vbguest winrm winrm-fs winrm-elevated
```

```
# Add some dependencies
sudo apt install git sshpass lftp rsync openssh-client python3-venv
```

```
git clone https://github.com/Orange-Cyberdefense/GOAD.git
cd GOAD
```

```
# verify installation
./goad.sh -t check -l GOAD -p virtualbox
```

```
# install
./goad.sh -t install -l GOAD -p virtualbox
```

```
# launch goad in interactive mode
./goad.sh
```

```
# The easy way is just launch ./goad.sh and use help ? in the interactive prompt
# Most used: install, status, start, stop, destroy, start_vm <vm_name>, etc.
```

Make sure you have enough **RAM**.

If you have any doubts, just use “**install**”.

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```
GOAD/virtualbox/local/192.168.56.X (bed9fa-goad-virtualbox) > ?

*** Manage Lab instance commands ***
status ..... show current status
start ..... start lab
stop ..... stop lab
destroy ..... destroy lab

*** Manage one vm commands ***
start_vm <vm_name> ..... start selected virtual machine
stop_vm <vm_name> ..... stop selected virtual machine
restart_vm <vm_name> ..... restart selected virtual machine
destroy_vm <vm_name> ..... destroy selected virtual machine

*** Extensions ***
list_extensions ..... list extensions
install_extension <extension> ..... install extension (providing + provisioning)
provision_extension <extension> ..... provision extension (provisioning only)

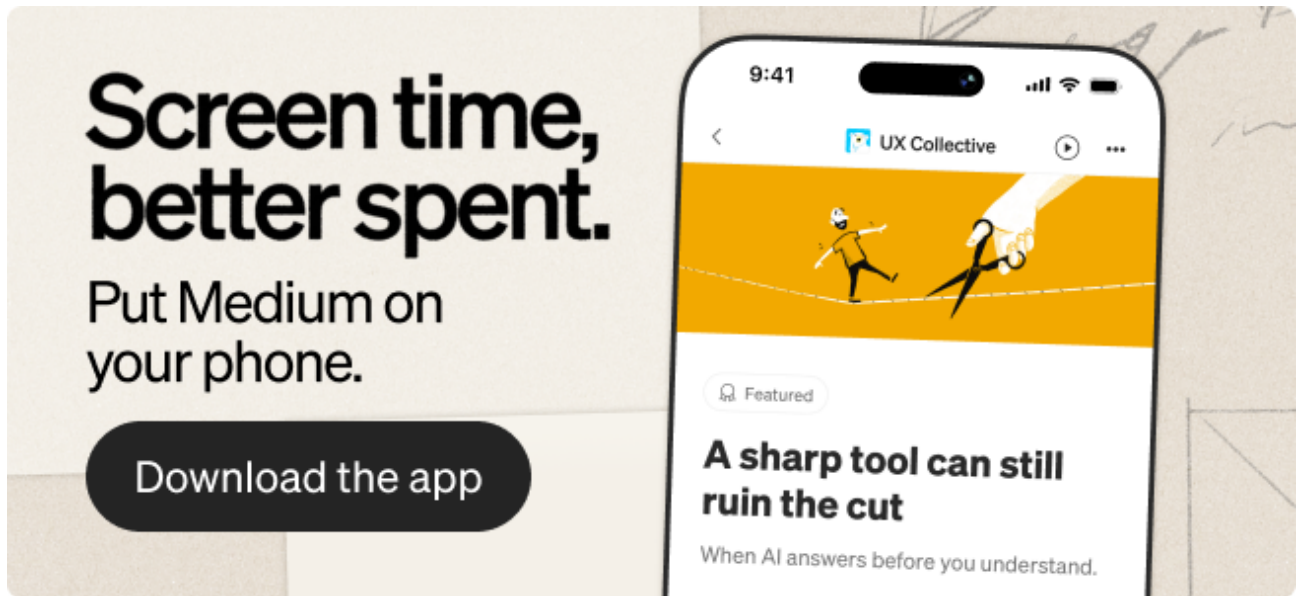
*** Providing (Vagrant/Terraform) ***
provide ..... run only the providing (vagrant/terraform) part

*** Provisioning (Ansible) ***
provision <playbook> ..... run specific ansible playbook
provision_lab ..... run all the current lab ansible playbooks
provision_lab_from <playbook> ..... run all the current lab ansible playbooks from <playbook>

*** Lab Instances ***
check ..... check dependencies before creation
install ..... install the current instance (providing + provisioning)
set_as_default ..... set instance as default
update_instance_files ..... update lab instance files
disable_vagrant ..... disable vagrant user
enable_vagrant ..... enable vagrant user
list ..... list lab instances
```

Options with “?”

It'll start downloading the VMs, just be patient.



Grab a coffee ☕

Read a book 📖

Watch an anime 🏯

Go for a run 🏃

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```
==> GOAD-DC01: Importing base box 'StefanScherer/windows_2019'...
==> GOAD-DC01: Matching MAC address for NAT networking...
==> GOAD-DC01: Checking if box 'StefanScherer/windows_2019' version '2021.05.15' is up to date
==> GOAD-DC01: Setting the name of the VM: GOAD-DC01
==> GOAD-DC01: Clearing any previously set network interfaces...
==> GOAD-DC01: Preparing network interfaces based on configuration...
GOAD-DC01: Adapter 1: nat
GOAD-DC01: Adapter 2: hostonly
==> GOAD-DC01: Forwarding ports...
GOAD-DC01: 5985 (guest) => 55985 (host) (adapter 1)
GOAD-DC01: 5986 (guest) => 55986 (host) (adapter 1)
GOAD-DC01: 22 (guest) => 2222 (host) (adapter 1)
==> GOAD-DC01: Running 'pre-boot' VM customizations...
==> GOAD-DC01: Booting VM...
==> GOAD-DC01: Waiting for machine to boot. This may take a few minutes...
GOAD-DC01: WinRM address: 127.0.0.1:55985
GOAD-DC01: WinRM username: vagrant
GOAD-DC01: WinRM execution_time_limit: PT2H
GOAD-DC01: WinRM transport: negotiate
==> GOAD-DC01: Machine booted and ready!
==> GOAD-DC01: Checking for guest additions in VM...
GOAD-DC01: The guest additions on this VM do not match the installed version of
GOAD-DC01: VirtualBox! In most cases this is fine, but in rare cases it can
GOAD-DC01: prevent things such as shared folders from working properly. If you see
GOAD-DC01: shared folder errors, please make sure the guest additions within the
GOAD-DC01: virtual machine match the version of VirtualBox you have installed on
GOAD-DC01: your host and reload your VM.
GOAD-DC01:
GOAD-DC01: Guest Additions Version: 6.1.22
GOAD-DC01: VirtualBox Version: 7.0
==> GOAD-DC01: Configuring and enabling network interfaces...
==> GOAD-DC01: Running provisioner: shell...
GOAD-DC01: Running: ../../../../vagrant/Install-WMF3Hotfix.ps1 as C:\tmp\vagrant-shell.ps1
==> GOAD-DC01: Running provisioner: shell...
GOAD-DC01: Running: ../../../../vagrant/ConfigureRemotingForAnsible.ps1 as C:\tmp\vagrant-she
```

Downloading and setting up the machines

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```
PLAY RECAP *****
dc01      : ok=11    changed=2    unreachable=0    failed=0
dc02      : ok=31    changed=14   unreachable=0    failed=0
dc03      : ok=8     changed=3    unreachable=0    failed=0
srv02     : ok=18    changed=10   unreachable=0    failed=0
srv03     : ok=12    changed=8    unreachable=0    failed=0

[*] Lab successfully provisioned in 22:43:43
GOAD/virtualbox/local/192.168.56.X (bed9fa-goad-virtualbox) > status
[*] CWD: /workspace/bed9fa-goad-virtualbox/provider
[*] Running command : vagrant status
Current machine states:

GOAD-DC01      running (virtualbox)
GOAD-DC02      running (virtualbox)
GOAD-DC03      running (virtualbox)
GOAD-SRV02     running (virtualbox)
GOAD-SRV03     running (virtualbox)

This environment represents multiple VMs. The VMs are all listed
above with their current state. For more information about a specific
VM, run `vagrant status NAME`.
```

Machines are all set up, use “status” to check if they’re running

Awesome, the labs are all installed and running.

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```
shanks@akagami:~/GOAD$ nxc smb 192.168.56.0/24
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10 / Server 2019
orth.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019
rth.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.12 445 MEEREEN [*] Windows Server 2016 Stand
ain:essos.local) (signing:True) (SMBv1:True)
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Stand
ain:essos.local) (signing:False) (SMBv1:True)
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10 / Server 2019
sevenkingdoms.local) (signing:True) (SMBv1:False)
Running nxc against 256 targets 100% 0:0
shanks@akagami:~/GOAD$
```

Check with nxc if you can reach them

Running **nxc smb 192.168.56.0/24** you’ll see that all the machines are working! 🙌

Really, that’s all of it. Just start hacking it. Have fun! 🤖

TROUBLESHOOTING

Usually you won't have problems, but if you do and happens to be about '**INTEL VT-x/EPT or AMD-V/RVI**' here's the solution:

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```
GOAD-DC01: 22 (guest) => 2222 (host) (adapter 1)
==> GOAD-DC01: Running 'pre-boot' VM customizations...
==> GOAD-DC01: Booting VM...
There was an error while executing 'VBoxManage', a CLI used by Vagrant
for controlling VirtualBox. The command and stderr is shown below.

Command: ["startvm", "734544e2-36fb-4d8e-98af-11449818249b", "--type", "headless"]

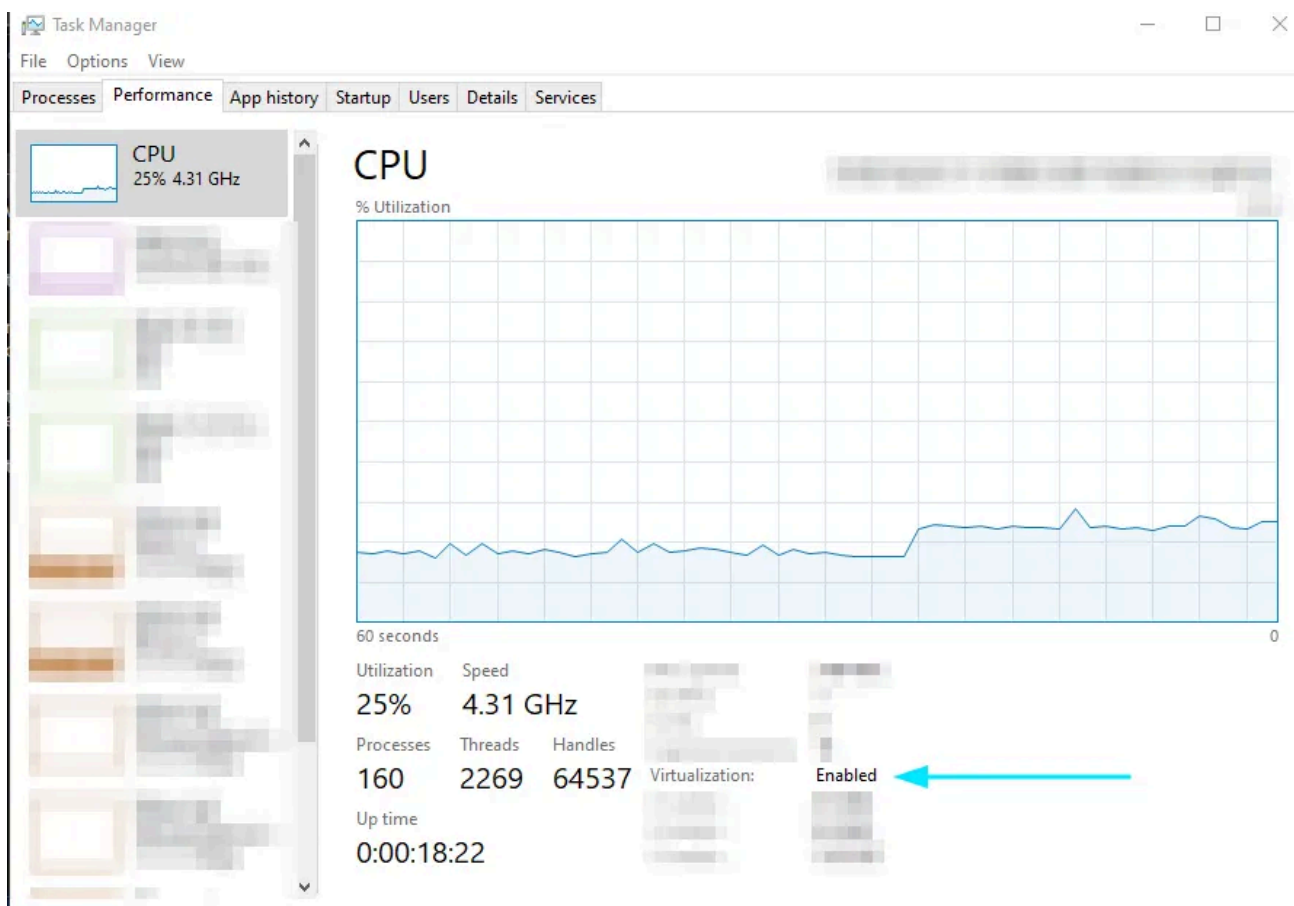
Stderr: VBoxManage: error: AMD-V is not available (VERR_SVM_NO_SVM)
VBoxManage: error: Details: code NS_ERROR_FAILURE (0x80004005), component ConsoleWrap, interface IConsole
[.] Providing error stop
```

AMD-V error

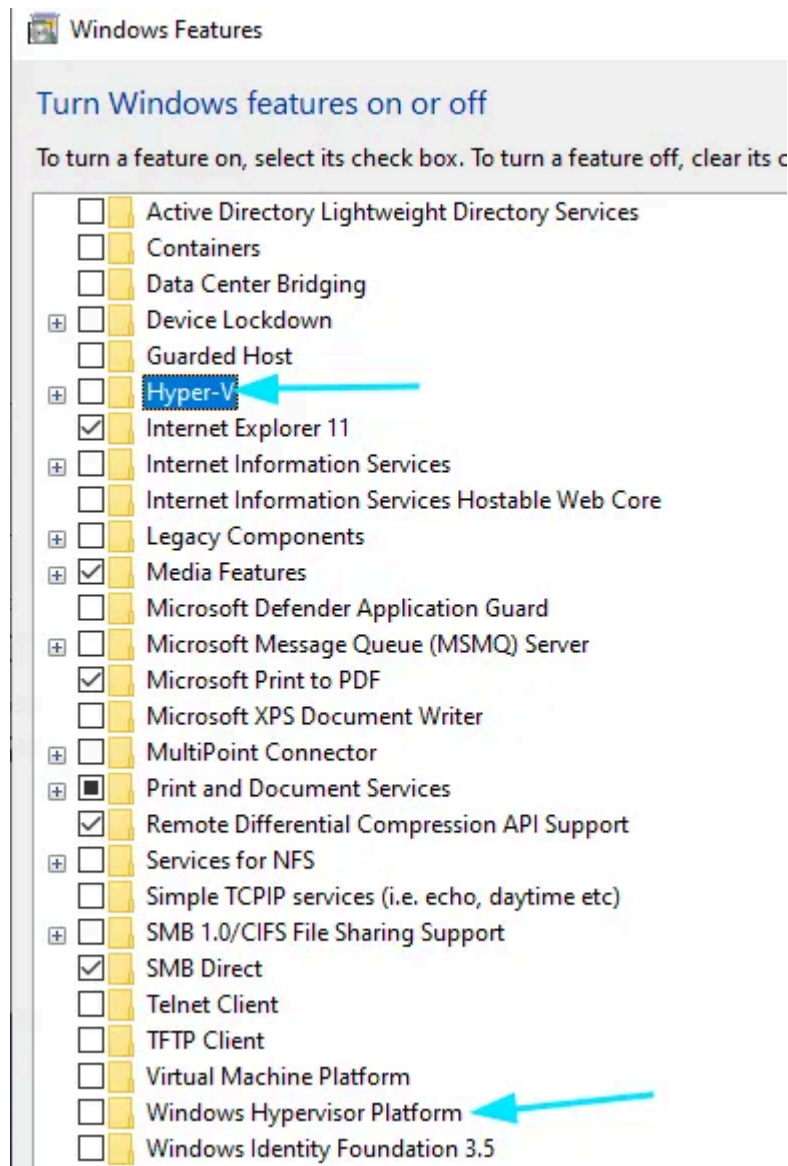
First of, you need to **enable Virtualization** in your **BIOS**. You can google search on how to do that for you motherboard.

You can check if you need to do this by going to your **CPU tab** using Task Manager like so:

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After enabling it, you need to **turn off any hypervisor** you have, you can do that by using the “**Turn Windows features on or off**”, cmd or Powershell if some doesn't work. Don't forget to **restart** your computer.



turn windows features on or off

```
C:\Users\Administrator>bcdedit /set hypervisorlaunchtype off
The operation completed successfully.
C:\Users\Administrator>
```

cmd option

`bcdedit /set hypervisorlaunchtype off`

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```
PS C:\Users\Administrator> Disable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-All
Do you want to restart the computer to complete this operation now?
[Y] Yes [N] No [?] Help (default is "Y"):
```

Powershell option

Disable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-All

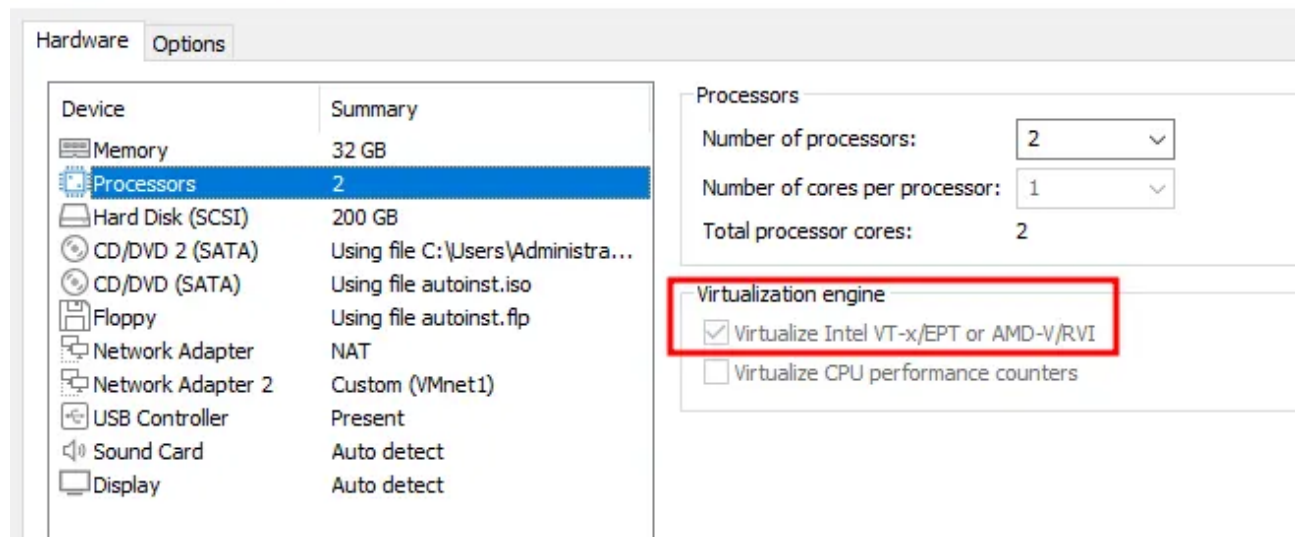
Go to your services and **stop/disable all related to Hyper-V.**

	HV Host Service	Manual (
	Hyper-V Data Exchange Service	Manual (
	Hyper-V Guest Service Interface	Manual (
	Hyper-V Guest Shutdown Service	Manual (
	Hyper-V Heartbeat Service	Manual (
	Hyper-V Host Compute Service	Manual (
	Hyper-V PowerShell Direct Service	Manual (
	Hyper-V Remote Desktop Virtualization Service	Manual (
	Hyper-V Time Synchronization Service	Manual (
	Hyper-V Virtual Machine Management	Automati
	Hyper-V Volume Shadow Copy Requestor	Manual (

disable all

Then, go to your Ubuntu and enable the option below.

Virtual Machine Settings



Enable the virtualize option

By doing those, you can start your Ubuntu normally and install GOAD.